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VERTEX DOMINATION IN MANGOLDT GRAPH

ABSTRACT

Domination theory of graphs is an important branch of Graph Theory and has many applications in Engineering, Communication Networks and many others. Recently Allan, R.B., and Laskar, R., [1,2], Cockayne, E.J., and Hedetniemi, S.T., [3], Haynes, T.W., and Slater, J.S., [4], Kulli, V.R., and Sigarkant, S.K., [5] et al. have studied the various domination parameters of graphs. Graphs associated with certain arithmetical functions which are usually called arithmetical graphs have been studied extensively by many researchers.

In this paper we introduce the concept of Mangoldt graph, associated with Mangoldt function $\Lambda(n)$, $n \geq 1$ is an integer and study its basic properties, vertex cover, neighbourhood set, dominating set and their parameters.

1. THE MANGOLDT GRAPH AND ITS PROPERTIES

Definition 1.1: Let $n \geq 1$ be an integer. The Mangoldt function $\wedge(n)$ is defined as follows.

$$\wedge(n) = \begin{cases} \log p & \text{, if } n = p^m \text{, for some prime } p \text{ and some } m \geq 1 \\ 0 & \text{, otherwise .} \end{cases}$$

The following is the table of the values of $\wedge(n)$, for $n = 1, 2, \ldots, 10$.

n :	1	2	3	4	5	6	7	8	9	10
$\wedge(n)$:	0	log2	log3	log2	log5	0	log7	log2	log3	0

Definition 1.2: Let $n \geq 1$ be an integer. The Mangoldt graph M_n is defined as the graph whose vertices are the elements of the set $\{1, 2, \ldots, n\}$ and two vertices x, y are adjacent (or (x, y) is an edge) if and only if $\wedge(x . y) = 0$. That is, the vertices x and y are adjacent if and only if x . y is not power of a prime.

Theorem 1.3 : Let $n \geq 2$ be an integer and let $p_1, p_2, p_3, \ldots, p_t$ be prime numbers $\leq$ n and let α_i be the largest positive integer such that $p_i^{\alpha_i} \leq n$. Then the number of edges in the Mangoldt graph M_n is given by :

$$^{n}C_2 - [(^{\alpha_1+1}C_2) + (^{\alpha_2+1}C_2) + \ldots\ldots\ldots + (^{\alpha_t+1}C_2)]$$

Proof: M_n has n vertices, namely $1, 2, \ldots, n$. The total number of edges that can be formed with n vertices is $^{n}C_2$. If p_i is any prime £ n, then there is no edge between any pair of vertices among $1, p_i, p_i^2, \ldots, p_i^{\alpha_i}$ in M_n. There are $(^{\alpha_i+1}C_2)$ such pairs from 1, . So the total number e of edges of M_n is given by

$$\varepsilon = {}^{n}C_2 - [(^{\alpha_1+1}C_2) + (^{\alpha_2+1}C_2) + \ldots\ldots\ldots + (^{\alpha_t+1}C_2)] .$$

Theorem 1.4: For $n \leq 5$, M_n is a disconnected graph.

Proof: Since the vertex 1 is not adjacent with other vertices, namely, 2, 3, 5, which are primes and 4 which is power of the prime 2 , that is , 2^2, M_n becomes a disconnected graph when $n \leq 5$.

Theorem 1.5 : For $n \geq 6$, the graph M_n is a connected graph .

Proof : Let $n \geq 6$ be an integer . If u is any vertex of M_n , $u \neq 1$, then clearly there is an edge between u and u + 1, since (u, u + 1) = 1. Also there is an edge between 1 and 6 . Thus M_n is a connected graph.

Theorem 1.6 : For $n \geq 6$, M_n is neither a bipartite graph nor a tree.

Proof : Let $n \geq 6$ be an integer. Then (2 3 5 2) is an odd cycle . Since odd cycles exist M_n is not bipartite . Also M_n is not a tree as it contains cycles.

Theorem 1.7 : For $n \geq 10$, M_n is Hamiltonian.

Proof : There is no edge between 1 and 2. But there always exists an edge between u and u+1 for all vertices u in M_n such that $2 \leq u \leq n$, since $(u, u+1) = 1$. Further there is an edge between 1 and 6, 6 and 2 and also between 5 and 7.

Case 1 : Suppose n is not a power of 2, but n is a prime or power of a prime, where n−1 is not a power of 2. Then it is clear that there is no edge between n and 1. But there is always an edge between n − 1 and 1, since n −1 is not a power of 2, and n −1 is a composite integer. Thus 2, 3, 4, 5, 7, ..., (n−1), 1, 6, n, 2 form a Hamilton cycle.

If n−1 is a power of 2, then there is no edge between n−1 and 1. Now n−1 being even, n−2 is odd and it is not a prime. Then there is an edge between n−2 and 1 and also between n − 1 and 6. Thus 2, 3, 4, 5, 7, 8,...., n−2, 1, 6, n−1, n, 2 form a Hamilton cycle.

Case 2 : Suppose n is a power of 2. Then n−2 is even and not a power of 2, so that there exists an edge between n − 2 and 1 and also between n and n − 2. Further n−3, n−1 being odd, whether they are prime or not there always exists an edge between n−3 and n − 1. Thus 2, 3, 4, 5, 7, 8, ..., n − 3, n − 1, n, n − 2, 1, 6, 2 form a Hamilton cycle.

Case 3: Suppose n is a composite integer. Obviously there is an edge between u and u+1 where u M_n, u1. We know that 1 and 6 are adjacent. Also 1 and n are adjacent, since $\wedge(1.n) = \wedge(n) = 0$. Then the Hamilton cycle is given by 2, 3, 4, 5, 6, 1, n, n−1, ..., 2.

2. VERTEX DOMINATION IN MANGOLDT GRAPH

Definition 2.1 : A set S of vertices which covers all the edges of a graph G is called **vertex cover** of G, in the sense that every edge of G is incident with some vertex in S. A minimum vertex cover is the one with minimum cardinality. The cardinality of a minimum vertex cover of a graph G is called the **covering number** and is denoted by b(G).

Theorem 2.2: Let $n > 2$ be any integer. Then for every prime $p \leq n$, the set

$$U = \{r / 1 < r \leq n, r \neq p^{\alpha}\}$$

$$= \{1, 2, \ldots, n\} - \{1, p, p^2, \ldots, p^{\alpha} / p^{\alpha} \leq n \text{ but } p^{\alpha+1} \nleq n\}$$

is a vertex cover of the Mangoldt graph M_n.

Proof: Let e be an edge of M_n. Then $e = (r, s)$, where $1 \leq r, s \leq n$, such that $\Lambda(r.s) = 0$. If one of r and s is in U, then there is nothing to prove. So let $r \notin U$ and $s \notin U$, so that $r = p^{\alpha}$ and $s = p^{\beta}$ for some $\alpha, \beta \geq 0$. Then $r.s = p^{\alpha+\beta}$, so that $\Lambda(r.s) = \log p \neq 0$, a contradiction. So at least one of r, s must belong to U, which implies that U is a vertex cover of M_n.

Corollary 2.3: The minimum vertex cover V_c of M_n is given by $V_c = \{1, 2, \ldots, n\} - \{1, 2, 2^2, \ldots, 2^t / 2^t \leq n \text{ but } 2^{t+1} \nleq n\}$ and its vertex covering number $\beta(M_n)$ is given by $\beta(M_n) = n - (t + 1)$, where t is a positive integer such that $2^t \leq n$ but $2^{t+1} \nleq n$.

Proof: Among all the sets $\{1, p, p^2, \ldots, p^{\alpha} / p^{\alpha} \leq n \text{ but } p^{\alpha+1} \nleq n\}$ where p is a prime number, the set $\{1, 2, 2^2, \ldots, 2^t / 2^t \leq n \text{ but } 2^{t+1} \nleq n\}$ is the set with maximum cardinality. So the set V_c is the set with minimum cardinality, so that V_c is a minimum vertex cover of M_n and $\beta(M_n) = n - (t + 1)$.

Definition 2.4: For a graph G, let V and E respectively denote the vertex set and edge set of G. For $v \in V$, the **neighbourhood** of a vertex v is defined as $N(v) = \{u \in V \mid (u,v) \in E\}$. The **closed neighbourhood** of v is given by $N[v] = N(v) \cup \{v\}$. A subset S of V is called a **neighbourhood set** of G, if $G = \bigcup_{v \in S} \langle N[v] \rangle$, where $\langle N[v] \rangle$ is the sub graph of G induced by N[v].

Definition 2.5: The cardinality of a minimum neighbourhood set of a graph G is called the **neighbourhood number** and it is denoted by $n_0(G)$.

Theorem 2.6: If $n > 5$, then for each positive integer $r \leq n$, which is not power of a prime, the singleton set $\{r\}$ is a minimum neighbourhood set of M_n and $n_0(M_n) = 1$.

Proof : Let $r \leq n$ be not a power of a single prime. Then

$$r = p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_t^{\alpha_t} \qquad \ldots\ldots\ldots\ldots (1)$$

where p_i's are primes and $\alpha_i \geq 0$, such that at least two α_i's are non - zero. If s is any vertex of M_n, then $r.s$ is not a power of a single prime. Hence $\Lambda(rs) = 0$ and s is adjacent to r. Thus the closed neighbourhood $N[r] = V = \{1, 2, 3, \ldots, n\}$, so that the subgraph $\langle N[r] \rangle$ of M_n induced by N[r] is M_n itself. This shows that $\{r\}$ is a neighbourhood set of M_n.

Clearly, the singleton set $\{r\}$ is a minimum neighbourhood set of M_n and $n_0(M_n) = 1$.

3. DOMINATING SETS AND INDEPENDENT SETS

Definition 3.1 : A **dominating set** of a graph G is a subset D of V such that each vertex of V-D is adjacent to at least one vertex of D. A minimum dominating set is the one with minimum cardinality. The **domination number** $\gamma(G)$ is the cardinality of a minimum dominating set.

Theorem 3.2 : If $n > 5$, then for each positive integer $r \leq n$ which is not a power of a single prime, the singleton set $\{r\}$ is a minimum dominating set of M_n and the dominating number $\gamma(M_n) = 1$.

Proof : The proof follows on the same lines as that of Theorem 2.6.

Remark 3.3: For the Mangoldt graph M_n the minimum neighbourhood set and minimum dominating set are the same.

Definition 3.4 : A subset D of V in a graph G is called a **total dominating set** of G if every vertex in V is adjacent to at least one vertex of D. The cardinality of a smallest total dominating set in G is called a total domination number of G and it is denoted by $\gamma_t(G)$.

Theorem 3.5 : If $n > 5$, then for each positive integer $r \leq n$, which is not a power of a single prime, the set $\{r, 1\}$ is a minimum total dominating set and the total domination number $\gamma_t(M_n) = 2$.

Proof : In Theorem 3.2, we have seen that for $n \geq 5$, the singleton set $\{r\}$, where r is not a power of a single prime is a minimum dominating set of M_n. So, r is adjacent to every s, $1 \leq s \leq n$, $s \neq r$.

Consider the set $\{1, r\}$. Since r is not a power of a single prime we have $\Lambda(1, r) = \Lambda(r) = 0$, so that r is adjacent to the vertex 1. Thus $\{1, r\}$ is a total dominating set of M_n. As $\gamma(M_n) = 1$, it follows that $\{1, r\}$ is a minimum total dominating set of M_n and hence $\gamma_t(M_n) = 2$.

Remark 3.6 : Since a number r which is not a power of a single prime is adjacent to every $s, r \neq s$, $1 \leq s \leq n$, we can even take the minimum total dominating set as $\{r, s\}$ instead of $\{r, 1\}$.

Definition 3.7 : A subset S of V is called an **Independent set** of a graph G if no two vertices of S are adjacent in G. The number of vertices in the largest independent set of a graph G is called the **independence number** and it is denoted by a (G). A single vertex of any graph G constitutes an independent set.

Theorem 3.8 : The independence number of M_n is $r + 1$, where r is the largest positive integer such that $2^r \leq n$.

Proof : For any prime $p \leq n$, consider the set

$$I_p = \{1, p, p^2, \ldots, p^r \mid p^r \leq n, \text{ but } p^{r+1} \nleq n\}.$$

Clearly, any two vertices in the above set are not adjacent to one another. So, it forms an independent set. Clearly, if p and q are primes $p < q \leq n$ then $|I_p| > |I_q|$

So, the set $\{1, 2, \ldots, 2^r \mid 2^r \leq n, \text{ but } 2^{r+1} \nleq n\}$ is the maximum independent set of M_n and the independence number of M_n is $r + 1$.

REFERENCES

[1] Allan, R. B., and Laskar, R., On domination and independent domination number of a graph , *Discrete Math* , 23(1978), 73-76.

[2] Allan, R. B., Lasker.R., and Hedetniemi.S.T., A note on total domination , *Discrete Math* , 49(1984) 7-13.

[3] Cockayne, E.J.,and Hedetniemi, S.T., Towards a theory of domination in graphs, *Networks*, 7(1977), 247-261

[4] Haynes, T.W., Hedetniemi, S.T., and Slater, J.S., *Fundamentals of Domination in Graphs*, Marcel Decker, Inc . (1998).

[5] Kulli, V.R., and Sigarkant, S.K., Further results on a neighbourhood number of a graph, *Indian J. Pure. Appl. Math.*, 23(1992), 575-577.

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